

Machine Learning: Advanced Techniques



I Generalization

I.2 Vapnik-Chervonenkis Theory

Werner Zellinger – Institute for Machine Learning

Copyright statement

This material, no matter whether in printed or electronic form, may be used for personal and non-commercial educational use only. Any reproduction of this material, no matter whether as a whole or in parts, no matter whether in printed or in electronic form, requires explicit prior acceptance of the authors.

Recap



AI Curriculum

- ML: Basic Techniques
Understand basic ML techniques
 - Classification, regression, uncertainty, regularization, ...
- ML: Supervised/Unsupervised Techniques
Use and choose ML techniques
 - supervised: NNs, SVMs, ...
 - unsupervised: clustering, projection, data reduction, ...
- **ML: Advanced Techniques**
Understand, design and modify ML techniques
 - generalization, inference, representation, dynamics

Content

- **Part I Generalization** When does learning work?
V. Vapnik, *The Nature of Statistical Learning Theory*, 2000.
- **Part II Inference** What is optimal under uncertainty?
H. Bishop, *Pattern Recognition and Machine Learning*, 2006.
- **Part III Representation** What functions/distributions can be expressed?
J. Tomczak, *Deep Generative Modeling*, 2022.
- **Part IV Learning dynamics** How do algorithms reach solutions?
S. Sra, S. Nowozin, and St. Wright, *Optimization for Machine Learning*, 2012.

Simple Learning Recipe

```
1  # data ~ p
2  X_train, y_train = load_data()
3  # part III representation: choose model class
4  model = Model(...)
5  # part II inference: define what should be estimated
6  objective = lambda theta: mean(loss(model(theta, X_train), y_train))
7  # part IV learning dynamics: compute estimator
8  theta_hat = Optimizer(...).minimize(objective)
9  # part I generalization: evaluate unseen performance
10 test_error = evaluate(model(theta_hat))
```

Part I: Generalization

I.1 Basics & necessity of assumptions

Risk, generalization, concentration, no free lunch

I.2 Vapnik-Chervonenkis theory

Consistency, uniform convergence, generalization bounds

I.3 Modern aspects

Double descent, distribution shifts

Content

- Motivation
- Formalization
 - Learnability (consistency)
 - Learning rate
- Estimation error
 - Approach by law of large numbers
 - Approach by uniform convergence
 - Generalization bound for finite classes
 - Generalization bound for infinite classes
 - Capacity measures beyond VC dimension
- Approximation error
 - Structural risk minimization

Motivation



Curse of Dimensionality

- Resolution grows exponentially with dimension (1961)
- Any $\hat{f}_l$ computed from samples $(\mathbf{x}_i, f(\mathbf{x}_i))_{i=1}^l$ and satisfying $\sup_{\mathbf{x} \in [0,1]^d} |\hat{f}_n(\mathbf{x}) - f(\mathbf{x})| \leq \varepsilon$ for all f in

$$\mathcal{F}(L) = \{f : [0, 1]^d \rightarrow \mathbb{R} \mid |f(\mathbf{x}) - f(\mathbf{x}')| \leq L \|\mathbf{x} - \mathbf{x}'\|_2\}$$

requires

$$l \geq C \left(\frac{L}{\varepsilon} \right)^d.$$



R. Bellman (1920–1984)

Empirical Observation

- Perceptron (1958)

$$\mathcal{F} = \{f : \mathbb{R}^d \rightarrow \mathbb{R} \mid f(\mathbf{x}) = \text{sign}(\mathbf{w}^\top \mathbf{x}), \mathbf{w} \in \mathbb{R}^d\}$$

- Observation: Models with $d \geq 1000$ work on novel data with $l \ll 2^{1000}$ samples.

$$O\left(\frac{L}{\epsilon}\right)^d \quad \text{versus} \quad O(d)$$

How can the perceptron work well on future data?



F. Rosenblatt (1928–1971)

Spoiler for this Lecture

- VC-dimension (1968) $VC(\mathcal{F})$
- If $VC(\mathcal{F}) < \infty$, then with prob. at least $1 - \delta$:

$$l \geq O\left(\frac{VC(\mathcal{F}) \log \frac{1}{\varepsilon} + \log \frac{1}{\delta}}{\varepsilon^2}\right),$$

where

$$\varepsilon = \sup_{f \in \mathcal{F}} |R(f) - R_{\text{emp}}(f)|.$$

Generalization scales with $VC(\mathcal{F})$, not with input dimension d .



V.N. Vapnik (1936–) & A.Y. Chervonenkis
(1938–2014)

Statistical Learning Theory

Q1 Under what conditions is learning possible?

Q2 How fast does learning occur?

Q3 How should we choose a function from data when the sample is small?

Q4 Which concrete algorithms implement these principles?

We will focus on binary classification (labels ± 1).

[V.N. Vapnik, *Statistical Learning Theory*, 1998]

Formalization



Empirical Risk Minimization

- Focus: Binary classifiers $f : X \rightarrow \{-1, 1\}$ and zero-one loss

$$L(y, y') = \mathbf{1}\{y \neq y'\} = \begin{cases} 0 & \text{if } y = y' \\ 1 & \text{if } y \neq y' \end{cases}$$

- Given: Dataset $\mathbf{Z} = (\mathbf{x}_i, y_i)_{i=1}^l$ i.i.d. drawn from $p(\mathbf{x}, y)$ on $X \times \{-1, 1\}$
- Goal: Find f with minimal **expected risk**

$$R(f) = \mathbb{E}_{(\mathbf{x}, y) \sim p}[L(y, f(\mathbf{x}))]$$

- Approach: Minimize **empirical risk**

$$R_{\text{emp}}(f) = \frac{1}{l} \sum_{i=1}^l L(y_i, f(\mathbf{x}_i))$$

Assumptions

- No parametric assumptions on p
- No distribution shift: p is unknown but fixed
- No realizability assumption: labels may be stochastic (Bayes noise allowed)
- Training samples are i.i.d. samples from p

Notation

- Bayes classifier f^* : Classifier with minimal expected risk
 - If we know p , we also know the Bayes classifier:

$$f^*(\mathbf{x}) = \begin{cases} +1 & \text{if } p(y = +1 | \mathbf{x}) > p(y = -1 | \mathbf{x}), \\ -1 & \text{if } p(y = -1 | \mathbf{x}) > p(y = +1 | \mathbf{x}) \end{cases}$$

- Learning aims to achieve risk close to $R(f^*)$
- Hypothesis class $\mathcal{F}$: Function class from which we select
- Best approximation $f_{\mathcal{F}}$: Optimal classifier in $\mathcal{F}$

$$f_{\mathcal{F}} \in \arg \min_{f \in \mathcal{F}} R(f)$$

- Learned classifier $\hat{f}$: Classifier chosen from $\mathcal{F}$ by some algorithm using $\mathbf{Z}$

Q1 Under what conditions is learning possible?

Definition (Convergence in Probability)

A sequence of random variables X_l converges in probability to X if

$$X_l \xrightarrow{P} X \iff \forall \varepsilon > 0 : \lim_{l \rightarrow \infty} \mathbb{P}(|X_l - X| > \varepsilon) = 0.$$

In learning:

$$\hat{f} = A(\mathbf{Z}), \quad \mathbf{Z} \sim p^l,$$

so $R(\hat{f})$ is a random variable.

Q1 Under what conditions is learning possible?

Definition (Consistency of Learning Algorithm)

Let $\hat{f} = A(\mathbf{Z})$ with $\mathbf{Z} \sim p^l$.

- $\mathcal{F}$ -consistent w.r.t. p if

$$R(\hat{f}) \xrightarrow{P} R(f_{\mathcal{F}}).$$

- Bayes consistent w.r.t. p if

$$R(\hat{f}) \xrightarrow{P} R(f^*).$$

- Universally $\mathcal{F}$ -/Bayes consistent if the above holds for all p .

Q1 $\iff$ Under what conditions does universally Bayes consistent algorithm exist?

Why universal consistency does not violate NFL

NFL (informal): $\forall A : \mathbf{Z} \mapsto \hat{f}, \forall l, \exists p : \mathbb{E}_{\mathbf{Z} \sim p^l} [R_p(A(\mathbf{Z}))] - R_p(f^*) \geq \frac{1}{2}$

Universal Bayes consistency: $\exists A : \mathbf{Z} \mapsto \hat{f}, \forall p : R_p(A(\mathbf{Z})) \xrightarrow{P} R_p(f^*)$

Key point

- NFL: for each **fixed** l , there exists a bad distribution p
- Consistency: for each **fixed** p , performance improves as $l \rightarrow \infty$

Conclusion

- No algorithm works for **all** p at fixed l
- One algorithm can work for every fixed p **as** $l \rightarrow \infty$

Q1 Under what conditions is learning possible?

$$R(\hat{f}) - R(f^*) = \underbrace{R(\hat{f}) - R(f_{\mathcal{F}})}_{\text{estimation error}} + \underbrace{R(f_{\mathcal{F}}) - R(f^*)}_{\text{approximation error}}$$

- Estimation error

- Effect of observing a finite sample instead of the full distribution
- Vanishes as $n \rightarrow \infty$ for $\mathcal{F}$ -consistent algorithms

- Approximation error

- Caused by restricting the search to $\mathcal{F}$
- Smaller for richer $\mathcal{F}$

$$Q1 \iff \text{universal Bayes consistency} \iff \begin{cases} \mathcal{F}\text{-consistency,} \\ \text{approximation error} = 0 \end{cases}$$

Q2 How fast does learning occur?

Q2 $\iff$ How fast goes $R(\hat{f}) \xrightarrow{P} R(f_{\mathcal{F}})$?

Definition (Learning Error)

Let $\hat{f} = A(\mathbf{Z})$, $\mathbf{Z} \sim p^l$. An algorithm achieves learning error $r(l, \delta)$ w.r.t. p and $\mathcal{F}$ if

$$\mathbb{P}\left(\left|R(\hat{f}) - R(f_{\mathcal{F}})\right| < r(l, \delta)\right) \geq 1 - \delta.$$

- Learning rate: how $r(l, \delta)$ decreases with l
- Sample complexity: minimal l^* such that $r(l^*, \delta) \leq \varepsilon$ (recall Perceptron experiments suggested l^* grows linearly with dimension)

Estimation Error



Estimation Error



Approach by Law of Large Numbers

Approaching Q1 by LLN

For empirical risk minimizer $\hat{f} \in \arg \min_{f \in \mathcal{F}} R_{\text{emp}}(f)$ it holds that

$$\begin{aligned} R(\hat{f}) - R(f_{\mathcal{F}}) &= R(\hat{f}) - R_{\text{emp}}(\hat{f}) + R_{\text{emp}}(\hat{f}) - R(f_{\mathcal{F}}) \\ &\leq R(\hat{f}) - R_{\text{emp}}(\hat{f}) + R_{\text{emp}}(f_{\mathcal{F}}) - R(f_{\mathcal{F}}) \end{aligned}$$

Theorem (Strong Law of Large Numbers)

Let $(X_i)_{i \geq 1}$ be i.i.d. with $\mathbb{E}[X_1] < \infty$. Then

$$\frac{1}{l} \sum_{i=1}^l X_i \xrightarrow[l \rightarrow \infty]{a.s.} \mathbb{E}[X_1].$$

- For fixed f : Choosing $X_i = L(y_i, f(x_i))$ gives $\mathbb{E}[X_i] = R(f)$ and $R_{\text{emp}}(f) \xrightarrow[l \rightarrow \infty]{a.s.} R(f)$.

Approaching Q2 by Hoeffding

- Idea: Quantify learning rate via concentration of $R_{\text{emp}}(f)$ around $R(f)$.

Theorem (Hoeffding)

If $(X_i)_{i=1}^l$ are i.i.d. and $X_i \in [a, b]$, then

$$\mathbb{P}\left(\left|\frac{1}{l} \sum_{i=1}^l X_i - \mathbb{E}[X_1]\right| \geq \varepsilon\right) \leq 2e^{-\frac{2l\varepsilon^2}{(b-a)^2}}.$$

- For fixed f : Choosing $X_i = L(y_i, f(\mathbf{x}_i)) \in [0, 1]$ gives

$$|R_{\text{emp}}(f) - R(f)| \leq \sqrt{\frac{\log(2/\delta)}{2l}} \quad \text{with probability at least } 1 - \delta.$$

space for deriving error bound from probabilistic statement

Why these approaches fail to prove consistency?

- For fixed f : $X_i = L(y_i, f(x_i))$ are i.i.d., LLN gives $R_{\text{emp}}(f) \xrightarrow{P} R(f)$.
- But, in a learning algorithm $\hat{f} = A(\mathbf{Z})$ with $\mathbf{Z} = \{(x_1, y_1), \dots, (x_l, y_l)\}$.
- Then, each

$$X_i = L(y_i, \hat{f}(x_i))$$

depends on the **entire dataset** used to construct $\hat{f}$.

- Hence $X_i = L(y_i, \hat{f}(x_i))$ are **not** independent.
- LLN and Hoeffding cannot directly be applied to prove consistency.

space for a counter example

Estimation Error



Approach by Uniform Convergence

Uniform convergence

For empirical risk minimizer $\hat{f} \in \arg \min_{f \in \mathcal{F}} R_{\text{emp}}(f)$ it holds that

$$\begin{aligned} R(\hat{f}) - R(f_{\mathcal{F}}) &= R(\hat{f}) - R_{\text{emp}}(\hat{f}) + R_{\text{emp}}(\hat{f}) - R(f_{\mathcal{F}}) \\ &\leq R(\hat{f}) - R_{\text{emp}}(\hat{f}) + R_{\text{emp}}(f_{\mathcal{F}}) - R(f_{\mathcal{F}}) \\ &\leq 2 \sup_{f \in \mathcal{F}} |R_{\text{emp}}(f) - R(f)| \end{aligned}$$

Definition (Uniform Convergence)

The empirical risk R_{emp} converges uniformly over $\mathcal{F}$ (w.r.t. p) to R if

$$\forall \varepsilon > 0 : \lim_{l \rightarrow \infty} \mathbb{P} \left(\sup_{f \in \mathcal{F}} |R_{\text{emp}}(f) - R(f)| > \varepsilon \right) = 0.$$

From uniform convergence to consistency

Lemma (Uniform convergence implies $\mathcal{F}$ -consistency)

Let $\hat{f} \in \arg \min_{f \in \mathcal{F}} R_{\text{emp}}(f)$ and any $\varepsilon > 0$,

$$\mathbb{P}\left(\left|R(\hat{f}) - R(f_{\mathcal{F}})\right| \geq \varepsilon\right) \leq \mathbb{P}\left(2 \sup_{f \in \mathcal{F}} |R_{\text{emp}}(f) - R(f)| \geq \varepsilon\right).$$

- Lemma: Uniform convergence implies $\mathcal{F}$ -consistency of ERM
- **VC fundamental theorem**: $\mathcal{F}$ -consistency of ERM implies uniform convergence

$$\text{Q1} \iff \text{universal Bayes consistency} \iff \begin{cases} \text{uniform convergence of } R_{\text{emp}} \\ \text{approximation error} = 0 \end{cases}$$

Estimation Error



Generalization Bound for Finite Function Classes

Finite function class (1)

- Apply Hoeffding to $X_i = L(y_i, f(x_i))$ with $\mathbb{E}[X_i] = R(f)$ for a fixed $f \in \mathcal{F}$. Since $X_i \in [0, 1]$, for any $\varepsilon > 0$ we obtain

$$\mathbb{P}(|R_{\text{emp}}(f) - R(f)| \geq \varepsilon) \leq 2e^{-2l\varepsilon^2}.$$

- **Finite hypothesis class.** Assume $|\mathcal{F}| = m$. Then

$$\mathbb{P}\left(\sup_{f \in \mathcal{F}} |R_{\text{emp}}(f) - R(f)| \geq \varepsilon\right) \leq 2m e^{-2l\varepsilon^2}.$$

- Hence with probability at least $1 - 2me^{-2l\varepsilon^2}$,

$$|R_{\text{emp}}(f) - R(f)| < \varepsilon \quad \text{for all } f \in \mathcal{F}.$$

Finite function class (2)

- It is often more convenient to express the bound in terms of a chosen failure probability δ . Setting $\delta = 2me^{-2l\epsilon^2}$ and solving for ϵ gives

$$\epsilon = \sqrt{\frac{\ln(2m) + \ln(1/\delta)}{2l}}.$$

Lemma (Generalization bound for finite hypothesis classes)

Let $|\mathcal{F}| = m$ and $0 < \delta < 1$. Then with probability at least $1 - \delta$, for all $f \in \mathcal{F}$,

$$R(f) \leq R_{\text{emp}}(f) + \sqrt{\frac{\ln(2m) + \ln(1/\delta)}{2l}}.$$

Finite function class: What does this result mean?

- The true risk is bounded by the empirical risk plus a term containing the capacity measure $\ln |\mathcal{F}|$
- If $|\mathcal{F}|$ is “small enough” compared to l (so that $\frac{\ln(2|\mathcal{F}|)}{l}$ is small), we obtain a small generalization bound
- The bound holds with probability $1 - \delta$; smaller δ leads to a looser bound

But

- If the function class is very large, the bound becomes useless
- If $|\mathcal{F}| = \infty$, the argument breaks completely

Estimation Error



Generalization Bound for Infinite Function Classes

Shattering coefficient

Definition (Shattering coefficient)

Define the restriction of the class $\mathcal{F}$ to the sample $x_1, \dots, x_l \in X$ as

$$\mathcal{F}_{x_1, \dots, x_l} = \{(f(x_1), \dots, f(x_l)) : f \in \mathcal{F}\} \subseteq \{-1, 1\}^l.$$

The *shattering coefficient* $\mathcal{N}(\mathcal{F}, l)$ is the maximal number of distinct vectors that can appear in $\mathcal{F}_{x_1, \dots, x_l}$ over all choices of $x_1, \dots, x_l \in X$:

$$\mathcal{N}(\mathcal{F}, l) = \max_{x_1, \dots, x_l \in X} |\mathcal{F}_{x_1, \dots, x_l}|.$$

- $X = \mathbb{R}$, $\mathcal{F} = \{1_{[a, \infty)} : a \in \mathbb{R}\}$
- $X = \mathbb{R}^2$, $\mathcal{F}$ = classifiers whose positive class is the region above a horizontal line

Generalization bound via shattering coefficient

Theorem

Let $\mathcal{F}$ be an arbitrary function class and $0 < \delta < 1$. Then with probability at least $1 - \delta$, all $f \in \mathcal{F}$ satisfy

$$R(f) \leq R_{\text{emp}}(f) + 2\sqrt{\frac{\ln(4\mathcal{N}(\mathcal{F}, 2l)) + \ln(1/\delta)}{l}}.$$

- **Capacity for infinite classes:** $\mathcal{N}(\mathcal{F}, l)$ instead of $|\mathcal{F}|$
- For uniform convergence (and $\mathcal{F}$ -consistency) we need $\frac{\ln \mathcal{N}(\mathcal{F}, 2l)}{l} \xrightarrow{l \rightarrow \infty} 0$

Proof idea

- Quantity to bound: $\sup_{f \in \mathcal{F}} |\mathbf{R}_{\text{emp}}(f) - \mathbf{R}(f)|$
- Difficulty: $\mathbf{R}(f) = \mathbb{E}[\mathbf{L}(f(X), Y)]$ depends on the unknown distribution
- Idea: approximate the expectation by the empirical risk on a second independent sample of size l
- Let $\mathbf{R}'_{\text{emp}}(f)$ denote this empirical risk (ghost sample)
- Symmetrization lemma (Vapnik):

$$\mathbb{P}\left(\sup_{f \in \mathcal{F}} |\mathbf{R}_{\text{emp}}(f) - \mathbf{R}(f)| > \varepsilon\right) \leq 2 \mathbb{P}\left(\sup_{f \in \mathcal{F}} |\mathbf{R}_{\text{emp}}(f) - \mathbf{R}'_{\text{emp}}(f)| > \frac{\varepsilon}{2}\right)$$

- Now the deviation depends only on the values of f on $2l$ sample points $\Rightarrow$ number of distinct cases $\leq \mathcal{N}(\mathcal{F}, 2l)$

Shattering and VC-dimension

Definition (Shattering)

A function class $\mathcal{F}$ **shatters** points $x_1, \dots, x_l$ if it can realize all 2^l possible labelings:

$$|\mathcal{F}_{x_1, \dots, x_l}| = 2^l.$$

Definition (VC-dimension)

The **VC-dimension** of $\mathcal{F}$ is the largest l such that some set of l points can be shattered by $\mathcal{F}$:

$$\text{VC}(\mathcal{F}) = \max \{l : \exists x_1, \dots, x_l \text{ shattered by } \mathcal{F}\}.$$

If no such maximum exists, $\text{VC}(\mathcal{F}) = \infty$.

Why VC-dimension?

Lemma (Vapnik–Chervonenkis–Sauer–Shelah)

Let $\mathcal{F}$ be a function class with $\text{VC}(\mathcal{F}) = d$. Then

$$\mathcal{N}(\mathcal{F}, l) \leq \sum_{i=0}^d \binom{l}{i} \leq \left(\frac{el}{d}\right)^d,$$

where the second inequality holds for $d \leq l$.

- $\text{VC}(\mathcal{F}) < \infty \Rightarrow \mathcal{N}(\mathcal{F}, l) = O(l^d)$
- $\text{VC}(\mathcal{F}) = \infty \Rightarrow \mathcal{N}(\mathcal{F}, l) = 2^l$ for arbitrarily large l

Examples

- $X = \mathbb{R}$, $\mathcal{F} = \{1_{[a,b]} : a < b\}$ (indicators of intervals)
- $X = \mathbb{R}^2$ (indicators of axis-aligned rectangles)
- $X = \mathbb{R}^2$ (indicators of convex polygons with d corners) $\Rightarrow VC(\mathcal{F}) = 2d + 1$
- $X = \mathbb{R}$, $\mathcal{F} = \{\text{sgn}(\sin(tx)) : t \in \mathbb{R}\} \Rightarrow VC(\mathcal{F}) = \infty$
- Hyperplanes with margin γ in $\mathbb{R}^r$ (data in ball of radius R):

$$VC(\mathcal{F}) = \min\left(r, \frac{2R^2}{\gamma^2}\right) + 1$$

space for VC-dimension of indicators

VC-generalization bound

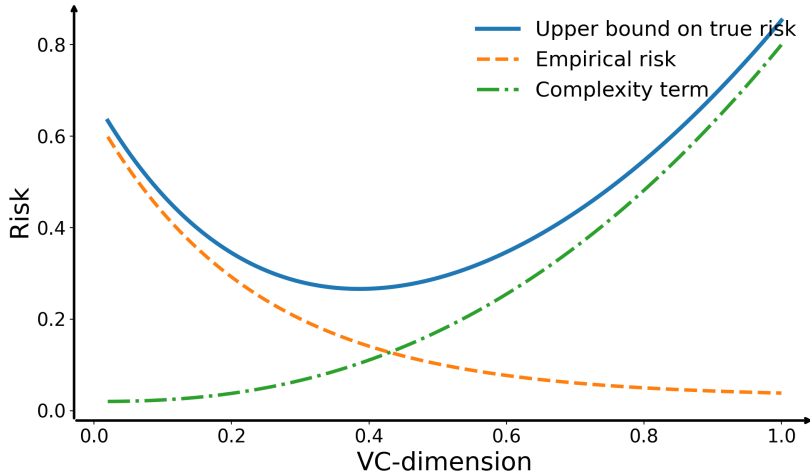
Theorem

Let $\mathcal{F}$ be a function class with $\text{VC}(\mathcal{F}) = d \leq l$. Then with probability at least $1 - \delta$, all $f \in \mathcal{F}$ satisfy

$$R(f) \leq R_{\text{emp}}(f) + 2\sqrt{\frac{d \ln\left(\frac{2el}{d}\right) + \ln(1/\delta)}{l}}.$$

Sample complexity (fix $\varepsilon = 2\sqrt{\frac{d \ln\left(\frac{2el}{d}\right) + \ln(1/\delta)}{l}}$, solve for l , ignore constants):

$$l^* = O\left(\frac{d}{\varepsilon^2}\right)$$



Where are we now?

- **Curse of dimensionality (Bellman):** function approximation requires $l \geq C(L/\varepsilon)^d$
- **Empirical observation:** perceptron suggests $O(d)$ samples
- **Using VC-dimension instead of d :** with high probability

$$\begin{aligned} R(\hat{f}) - R(f^*) &= \underbrace{R(\hat{f}) - R(f_{\mathcal{F}})}_{\text{estimation}} + \underbrace{R(f_{\mathcal{F}}) - R(f^*)}_{\text{approximation}} \\ &\leq 2 \sup_{f \in \mathcal{F}} |R_{\text{emp}}(f) - R(f)| + R(f_{\mathcal{F}}) - R(f^*) \\ &\leq O\left(\sqrt{\frac{VC(\mathcal{F}) \log l + \log(1/\delta)}{l}}\right) + \underbrace{R(f_{\mathcal{F}}) - R(f^*)}_{\text{how to control?}} \end{aligned}$$

Estimation Error



Capacity Measures Beyond VC-Dimension

Why go beyond VC-dimension?

- VC-dimension provides a **distribution-free** capacity measure for **binary classification**
- However, modern learning problems often involve
 - margin-based predictors
 - real-valued function classes
 - data-dependent generalization analysis
- In these settings, alternative notions of **capacity** are needed.

Examples of capacity measures

- **Rademacher complexity**: How well functions in $\mathcal{F}$ correlate with random labels.

$$\mathfrak{R}_l(\mathcal{F}) = \mathbb{E}_{\sigma_1, \dots, \sigma_l} \left[\sup_{f \in \mathcal{F}} \frac{1}{l} \sum_{i=1}^l \sigma_i f(x_i) \right],$$

where $\sigma_i \in \{-1, 1\}$ are independent random signs.

- **Covering numbers**: Minimal number of balls of radius ε needed to cover $\mathcal{F}$.

$$\mathcal{N}(\mathcal{F}, \varepsilon, \|\cdot\|) = \min \left\{ N \in \mathbb{N} : \exists f_1, \dots, f_N \in \mathcal{F} \text{ s.t. } \forall f \in \mathcal{F}, \min_{1 \leq j \leq N} \|f - f_j\| \leq \varepsilon \right\}.$$

Measures the *geometric size* of $\mathcal{F}$.

- **Fat-shattering dimension**: Largest number of points that can be separated with margin γ by functions in $\mathcal{F}$. Scale-sensitive extension of VC-dimension.

Approximation Error



Approximation Error

- Recall

$$R(\hat{f}) - R(f^*) = \underbrace{R(\hat{f}) - R(f_{\mathcal{F}})}_{\text{estimation}} + \underbrace{R(f_{\mathcal{F}}) - R(f^*)}_{\text{approximation}}$$

- To reduce the **approximation error**, the function class should become richer
- Idea: use a **nested sequence of function classes**

$$\mathcal{F}_1 \subset \mathcal{F}_2 \subset \mathcal{F}_3 \subset \dots$$

- Few samples $\Rightarrow$ small class (avoid overfitting)
- Many samples $\Rightarrow$ larger class (smaller approximation error)

Structural Risk Minimization (SRM)

- Choose the complexity level $k(l)$ depending on the sample size:

$$\hat{f}_l \in \arg \min_{f \in \mathcal{F}_{k(l)}} R_{\text{emp}}(f)$$

- Sufficient conditions for **universal Bayes consistency** of SRM:

$$\inf_{f \in \mathcal{F}_{k(l)}} R(f) \xrightarrow{l \rightarrow \infty} R(f^*) \quad \text{and} \quad \frac{VC(\mathcal{F}_{k(l)}) \log l}{l} \xrightarrow{l \rightarrow \infty} 0$$

- Interpretation:
 - classes grow rich enough to make approximation error vanish
 - classes grow slowly enough to keep estimation error under control

Summary



Summary

- **Curse of dimensionality (Bellman):** approximation requires $l \geq C(L/\varepsilon)^d$
- **Empirical observation:** perceptron suggests $O(d)$ samples
- **Using VC-dimension instead of d :** with high probability

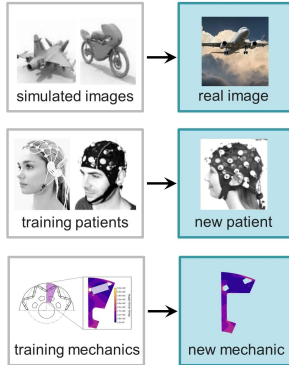
$$\begin{aligned} R(\hat{f}) - R(f^*) &= \underbrace{R(\hat{f}) - R(f_{\mathcal{F}})}_{\text{estimation}} + \underbrace{R(f_{\mathcal{F}}) - R(f^*)}_{\text{approximation}} \\ &\leq O\left(\sqrt{\frac{VC(\mathcal{F}) \log l + \log(1/\delta)}{l}}\right) + R(f_{\mathcal{F}}) - R(f^*) \end{aligned}$$

- **Structural Risk Minimization (SRM):** control the approximation error $R(f_{\mathcal{F}_{k(l)}}) - R(f^*)$ by nested spaces $\mathcal{F}_1 \subset \mathcal{F}_2 \subset \dots$ and choose complexity $k(l)$ depending on l to balance estimation and approximation error

Next topics



Distribution shifts



nature

Explore content ▾ About the journal ▾ Publish with us ▾

[nature](#) > [review articles](#) > [article](#)

Review | Published: 02 August 2023

Scientific discovery in the age of artificial intelligence

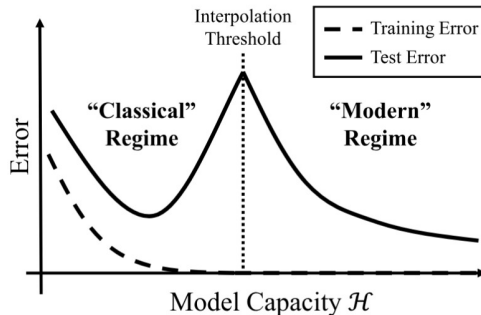
[Hanchen Wang](#), [Tianfan Fu](#), [Yuanqi Du](#), [Wenhao Gao](#), [Kexin Huang](#), [Ziming Liu](#), [Payal Chandak](#), [Shengchao Liu](#), [Peter Van Katwyk](#), [Andreea Deac](#), [Anima Anandkumar](#), [Karianne Bergen](#), [Carla P. Gomes](#), [Shirley Ho](#), [Pushmeet Kohli](#), [Joan Lasenby](#), [Ramsundar](#), [Le Song](#), [Jimeng](#)

Nature **620**, 47–60 (2023) | C

“The question of **out-of-distribution generalization is at the frontier of AI** research. A neural network trained on data from a specific regime may discover regularities that do not generalize in a different regime whose underlying **distribution has shifted** (Box 1)”

Double descent

model	# params	random crop	weight decay	train accuracy	test accuracy
Inception	1,649,402	yes	yes	100.0	89.05
		yes	no	100.0	89.31
		no	yes	100.0	86.03
		no	no	100.0	85.75
(fitting random labels)		no	no	100.0	9.78
Inception w/o BatchNorm	1,649,402	no	yes	100.0	83.00
		no	no	100.0	82.00
		no	no	100.0	10.12
Alexnet	1,387,786	yes	yes	99.90	81.22
		yes	no	99.82	79.66
		no	yes	100.0	77.36
		no	no	100.0	76.07
(fitting random labels)		no	no	99.82	9.86
MLP 3x512	1,735,178	no	yes	100.0	53.35
		no	no	100.0	52.39
		no	no	100.0	10.48
MLP 1x512	1,209,866	no	yes	99.80	50.39
		no	no	100.0	50.51
		no	no	99.34	10.61



[Z Chiyuan et al. Understanding deep learning (still) requires rethinking generalization, *Comm ACM* 2021]

[Ch Vladimir and E Lee. To understand double descent, we need to understand VC theory, *Neural Networks* 2024]

Werner Zellinger

Institute for Machine Learning



☎ +43 732 2468 4532
✉ @zellinger@ai-lab.jku.at
🌐 jku.at/ai-lab-team